Shanker Valipireddy

Principal Architect / Associate Director

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Professional Summary

Principal Architect and applied-AI leader with 20+ years of experience taking AI systems from discovery through pilot, stabilization, and production rollout in enterprise environments. Currently **Principal Architect / Associate Director** at Aflac, driving enterprise GenAI and agentic AI adoption across 100+ engineering teams. Hands-on builder of multi-agent systems (LangGraph, CrewAI, MCP, LangChain), enterprise LLM evaluation and governance, and cloud-scale deployment on AWS, Azure, and GCP.

Work Experience

Jan 2025 - Present

Aflac *Principal Architect / Associate Director — AI/ML, Agentic AI, GenAI* *Georgia, USA*

- Claims Automation – From uploading of claim forms to Adjudication automation using Agentic AI, ML, NLP,NER, Knowledge Graph
- Spearheaded enterprise AI enablement: drove GenAI and agentic AI adoption across 100+ engineering teams via role-specific workshops and standardized ‘AI Playbooks’.
- Led migration of a legacy $3M/year Cloud Ingestion/ETL platform to cloud-native AWS (Airflow, Glue), using AI agents to refactor complex pipelines and modernize enterprise applications.
- Deployed an agentic code-generation pipeline (AWS Kiro agents) converting ~20k data pipelines across 82 business domains into Airflow DAGs and Glue jobs — 99.3% QA pass rate.
- Implemented AI governance metrics (cycle time, defect-escape rates, token efficiency) and Human-in-the-Loop guardrails for production quality, security, and compliance.

Jan 2023 - Jan 2025

JPMorgan Chase *Principal Architect – AI/ML, Agentic AI, GenAI* *Tampa, Florida, USA*

- Designed multi-agent systems with LangGraph, CrewAI, MCP, and LangChain agents with LLM structured outputs for reasoning, tool use, and external interaction.
- Built Generative AI proof-of-concepts including regulatory text extraction (JPMC RCMA) on GCP/Vertex AI with GPT, Claude, Llama, and Mistral models.
- Established LLM fine-tuning and MLOps workflows (Llama 2/3.1, Mistral, Cohere, GPT-4) and transformer NLP (RoBERTa, DistilBERT) for text analytics and automation.

Jan 2016 - Jan 2023

Western Digital *Senior Principal / Technologist – AI/ML, Big Data* *California, USA*

- Designed an AI/ML data platform on composable infrastructure and Kubernetes, delivered as on-demand SaaS for edge-to-cloud AI use cases (video/image analytics, text intelligence).
- Delivered data-lakehouse components (MongoDB, Cassandra, Delta Lake), zoned-storage MySQL solutions, and smart archiving/analytics prototypes on OpenStack.

- Drove inference and data-pipeline performance optimization across GPUs, RDMA/RoCE, NVMe-oF, computational storage, and object storage.

Oct 2014 - Dec 2015 — **Cisco Systems, Inc.** *Principal Architect – Big Data, AI/ML, NLP (Contract)* — *California, USA*

- Built deep-learning anomaly detection over network-device telemetry using pre-trained embeddings and clustering over log semantics and context.
- Integrated live devices for real-time detection with an operator dashboard and a human feedback loop for continuous model refinement.

Oct 2010 - Oct 2014 — **Franklin Templeton** *Sr. Java Developer* — *California, USA*

- Built Hadoop/HBase ETL and MapReduce analytics; Hive/Pig reporting and Oracle integration via Sqoop; Twitter sentiment analysis served through the HBase REST API.

Jun 2003 - Oct 2010 — **IBM, Wipro & CF India** *Java Engineer* — *Dublin, CA, USA*

- Enterprise application integration (Java/Spring, TIBCO); proof-of-concept design and development.

Education

Jan 1999 - Jun 2003 — **B.Tech, Computer Science and Engineering**

Patents

Method and Apparatus for Smart Archiving and Analytics — US 10,360,193, issued July 23, 2019.
Intelligent Data Access Across Tiered Storage Systems — US 11,544,216, issued January 3, 2023.

Publications & Conference Talks

Enabling an Optimal Application Ecosystem on ZNS Storage. *Flash Memory Summit, Santa Clara, CA, August 2022.*
An Artificial Intelligence Data Pipeline for Storing and Processing Ingested Data. *Flash Memory Summit, Santa Clara, CA, 2019.*
Flash Storage Complementing a Data Lake for Real-time Insight. *Flash Memory Summit, Santa Clara, CA, 2018.*
An Analytics System on OpenStack for Manufacturing. *Western Digital corporate white paper, 2018.*
An Analytics System for Optimizing Manufacturing Processes. *Western Digital corporate white paper, 2018.*
Use Cases for ActiveScale Data Pipeline Service. *Western Digital corporate white paper, 2019.*
ActiveScale Data Pipeline Service with Elasticsearch — Best Practices for Metadata Indexing and Search. *Western Digital corporate white paper, 2019.*
AI Solution — Object Storage Content Analysis. *Corporate presentation, Western Digital, 2018.*
AI/ML MLPerf/TensorFlow/PyTorch Performance Benchmark — Image Detection. *Corporate presentation, Western Digital, 2018.*
AI/ML/Deep Learning (TensorFlow/Keras) for Analyzing Fraud Detection. *Presentation, 2019.*

Certifications

AWS Certified Machine Learning — Specialty; Microsoft Azure AI Engineer (AZ-102); Generative AI with Large Language Models; Machine Learning, NLP, and Deep Learning specializations; Transformer Models and BERT; LLMs: Application through Production; Building Conversational AI Applications (NVIDIA); Responsible AI; AI in the Data Center (NVIDIA); Apache Cassandra Certified Developer; MongoDB Certified Developer; Spark ML; Big Data XSeries (UC BerkeleyX); Advanced Predictive Analytics using R (IIT Hyderabad); PMP Training.

Awards

IBM Bravo Award.
Best Employee Award.